

PREMKUMAR M. PATEL

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EDUCATION

Sankalchand Patel University

Visnagar, Gujarat, India

B.Tech in Computer Engineering, CGPA: 7.80 / 10.0

2023 – 2027

- Coursework: Machine Learning, Artificial Intelligence, Deep Learning Applications, Database Management Systems, Operating Systems, Data Structures & Algorithms, Software Engineering, NLP, Compiler Design, GenAI
- Organizer in the college technical club, managing hackathons and technical workshops

TECHNICAL SKILLS

Languages: Python, C++, SQL, TypeScript

Backend & APIs: FastAPI, Flask, RESTful APIs, Async I/O, Pydantic, Webhooks, HMAC Verification, Rate Limiting

AI & Agentic Systems: LangGraph, LangChain, Claude API, OpenAI API, RAG Pipelines, Tool/Function Calling, Prompt Engineering

Databases & Vector Search: PostgreSQL, Redis, SQLite, FAISS, Qdrant, Pinecone

System Design & Scalability: Docker, Microservices, Message Queues (Celery), Load Balancing, Caching Strategies, API Gateways, CI/CD, Nginx

DevOps & Tools: Git, GitHub, Linux/Bash, Vercel, Render, Postman

FEATURED PROJECTS

AgentPay

Aug 2026

Python, FastAPI, LangGraph, PostgreSQL, Redis, Next.js, Razorpay API

[Live Demo](#) | [GitHub](#)

- Built an autonomous buyer-agent system where the LLM proposes purchases and a separate deterministic policy engine gates every payment before it reaches the Razorpay API
- Logged every state change in an append-only audit trail backed by database triggers, with merchant data isolated per tenant and hard spend limits per consumer
- Set up a sandbox-to-production pipeline with webhook-driven agent integration, HMAC signature verification, and automatic retry with backoff

CampusCircle

Nov 2025 – Jan 2026

Next.js, FastAPI, PostgreSQL, LLaMA 3.3 70B, TypeScript

[Live Demo](#) | [GitHub](#)

- Designed and deployed the full backend and AI infrastructure for a verified student platform, scaling it solo to over 200 active campus users with average API response times under 150ms
- Shipped three production AI features on LLaMA 3.3 70B: an automated support agent ("Reva"), an adaptive quiz engine that adjusts question difficulty in real time, and a YouTube-to-study-notes summarizer
- Placed in the top 18 of over 500 participants nationwide in The Maverick Effect AI Challenge, run by the Dewang Mehta Foundation Trust

CliniqueAI

Jan 2026 – Feb 2026

Python, Flask, PostgreSQL, Scikit-learn, XGBoost, LangChain, Pydantic

[Live Demo](#) | [GitHub](#)

- Trained and benchmarked Logistic Regression, Random Forest, and XGBoost classifiers for clinical diabetes risk prediction, tuning the decision boundary for explainability and high recall
- Built a RAG-based patient copilot grounded on localized EHR records, with transactional CRUD endpoints, Pydantic data schemas, and physician notification alerts
- Removed cold-start latency on Render's free tier by setting up an automated heartbeat with UptimeRobot

HONORS

National AI Hackathon (Unstop) — Top 20 Finalist, out of 1,600+ teams

National Level

Selected among the top 20 software teams nationwide for CliniqueAI, a diagnostic copilot

Feb 2026

The Maverick Effect AI Challenge — Top 18 Finalist, out of 500+ participants

National Level

Recognized by the Dewang Mehta Foundation Trust for CampusCircle, an AI community platform

2026

CERTIFICATIONS

HackerRank Software Engineer Certification

2025

Problem Solving, SQL, REST APIs